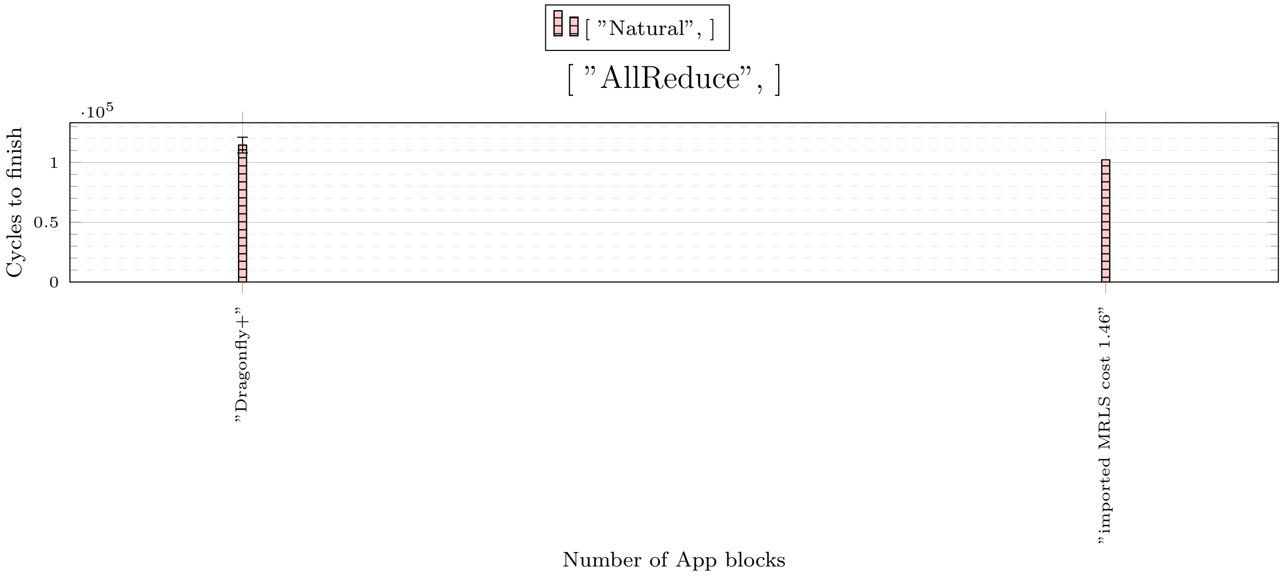

Figure 1: X: ["AllReduce",]

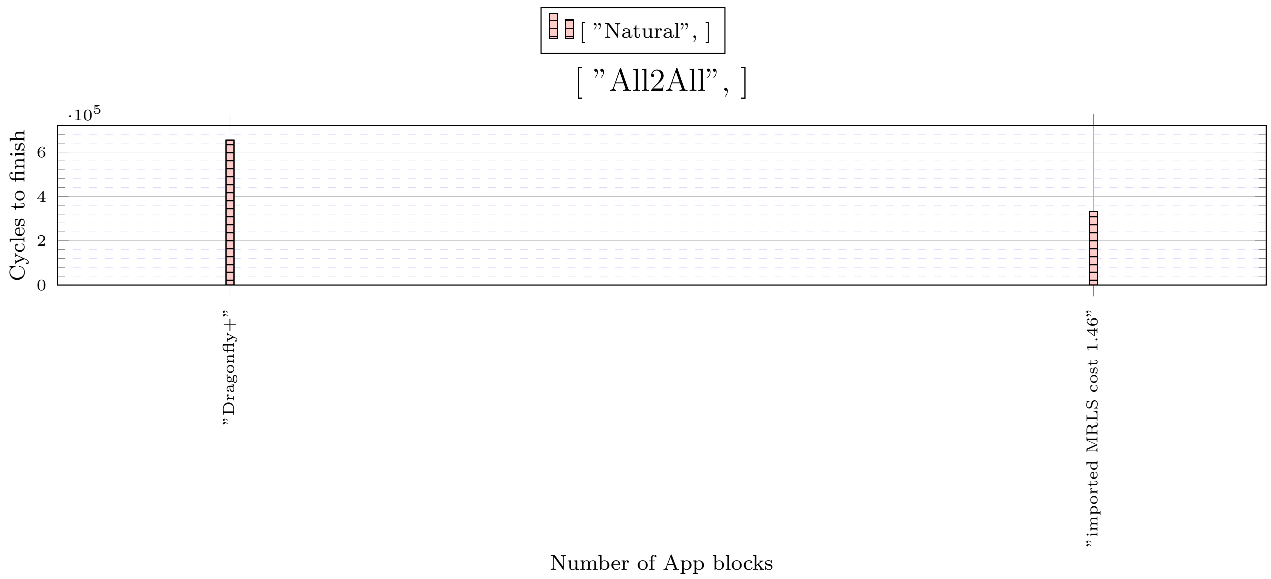

Figure 2: X: ["All2All",]

The following versions used in the simulations.

- heads/alex-dev-defe967c1db10e3b873fe074f274b6bdc9c42bb8(0.6.3)
- heads/master-dirty-ce88a576295ce68f40736599a9e3e04f24f38801(0.6.3)


```
Configuration{
  random_seed: ![ 42, 935, 128643 ],
  warmup: 20000,
  measured: 10000000000000000,
  general_frequency_divisor: 2,
  topology: topologia![
    MultiStage{
      stages: [
        ExplicitStageFile{ filename: "../../MRLS_1280_19_760_32_a_net" }],
      servers_per_leaf: 13,
      legend_name: "imported MRLS cost 1.46"},
    Megafly{ global_ports_per_spine: 16, servers_per_leaf: 16, global_arrangement: Palmtree, group_size: 16, number_of_groups: 65, legend_name:
"Dragonfly+" }],
  traffic: ![
    TrafficMap{
      tasks: 16640,
      map: ![
        Composition{
          patterns: [
            Identity{ legend_name: "Natural" },
            CartesianEmbedding{
              source_sides: [ 16384 ],
              destination_sides: [ 16640 ]}],
          middle_sizes: [ 16384, 16640 ],
          legend_name: "Natural"}],
      application: AllReduce{ tasks: 16384, data_size: 16384 },
      legend_name: "AllReduce"},
    TrafficMap{
      tasks: 16640,
      map: ![
        Composition{
          patterns: [
            Identity{ legend_name: "Natural" },
            CartesianEmbedding{
              source_sides: [ 16512 ],
              destination_sides: [ 16640 ]}],
          middle_sizes: [ 16512, 16640 ],
          legend_name: "Natural"}],
      application: ![
        All2All{ tasks: 16512, data_size: 132096 }],
      legend_name: "All2All"}],
  maximum_packet_size: 16,
  router: InputOutput{
    virtual_channels: 4,
    virtual_channel_policies: topologia![
      [
        MapHop{
          hop_to_policy: [
            ArgumentVC{
              allowed: [ 0 ]},
            ArgumentVC{
              allowed: [ 0 ]},
            ArgumentVC{
              allowed: [ 1 ]},
            ArgumentVC{
              allowed: [ 1 ]},
            ArgumentVC{
              allowed: [ 2 ]},
            ArgumentVC{
              allowed: [ 2 ]},
            ArgumentVC{
              allowed: [ 3 ]},
            ArgumentVC{
              allowed: [ 3 ]}]},
        VecLabel{
          label_vector: [ 0, 64, 80 ]},
        OccupancyFunction{ label_coefficient: 1, occupancy_coefficient: 1, product_coefficient: 0, constant_coefficient: 0, use_internal_space: true,
use_neighbour_space: true, aggregate: true },
        OccupancyFunction{ label_coefficient: 1, occupancy_coefficient: 1, product_coefficient: 0, constant_coefficient: 0, use_internal_space: true,
use_neighbour_space: true, aggregate: false },
        LowestLabel,
        EnforceFlowControl,
        Random],
      [
        MapHop{
          hop_to_policy: [
            MapLabel{
              label_to_policy: [
                Chain{
                  policies: [
                    OccupancyFunction{ label_coefficient: 0, occupancy_coefficient: 1, product_coefficient: 0, constant_coefficient: 0,
use_internal_space: true, use_neighbour_space: true, aggregate: false }]},
                Chain{
                  policies: [
                    AverageOccupancyFunction{
                      label_coefficient: 0,
                      occupancy_coefficient: 2,
                      product_coefficient: 0,
                      constant_coefficient: 16,
                      use_internal_space: true,
                      use_neighbour_space: true,
                      exclude_minimal_ports: false,
                      virtual_channels: [ 0 ]}]}]},
            MapLabel{
              label_to_policy: [
                Chain{
                  policies: [
                    OccupancyFunction{ label_coefficient: 0, occupancy_coefficient: 1, product_coefficient: 0, constant_coefficient: 0,
```

use_internal_space: true, **use_neighbour_space**: true, **aggregate**: false }]},
Chain{
policies: [
AverageOccupancyFunction{
label_coefficient: 0,
occupancy_coefficient: 3,
product_coefficient: 0,
constant_coefficient: 16,
use_internal_space: true,
use_neighbour_space: true,
exclude_minimal_ports: false,
virtual_channels: [0],
exclude_link_classes: [0]}]}]}],
above_policy: *MapLabel*{
label_to_policy: [
Chain{
policies: [
OccupancyFunction{ **label_coefficient**: 0, **occupancy_coefficient**: 1, **product_coefficient**: 0, **constant_coefficient**: 0, **use_internal_space**: true, **use_neighbour_space**: true, **aggregate**: false }]},
Chain{
policies: [
OccupancyFunction{ **label_coefficient**: 0, **occupancy_coefficient**: 1, **product_coefficient**: 0, **constant_coefficient**: 48, **use_internal_space**: true, **use_neighbour_space**: true, **aggregate**: false }]}]}},
LowestLabel,
EnforceFlowControl,
Random]],
allocator: *Random*,
buffer_size: 128,
bubble: false,
flit_size: 16,
intransit_priority: false,
allow_request_busy_port: true,
output_buffer_size: 64,
crossbar_frequency_divisor: 1,
crossbar_delay: 2},
routing: **topologia![**
Polarized{ **include_labels**: true, **strong**: false, **legend_name**: "Polarized" },
MegaflyAD{
first_allowed_virtual_channels: [0, 1],
second_allowed_virtual_channels: [2, 3],
minimal_to_deroute: [1, 1, 0],
source_group_pattern: [
[*None*, *None*],
[*None*, *None*]],
intermediate_group_pattern: [
[*None*, *None*],
[*None*, *None*]],
destination_group_pattern: [
[*None*, *None*],
[*None*, *None*]],
global_pattern_per_hop: [
[*None*, *None*],
[*None*, *None*]],
legend_name: "FPAR 2 vc"}],
link_classes: [
LinkClass{ **delay**: 2 },
LinkClass{ **delay**: 2 },
LinkClass{ **delay**: 2 }],
launch_configurations: [
Slurm{ **job_pack_size**: 1, **time**: "3-00:00:00" }]}